# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** second | 1 minute = 60 seconds. |
| 2 | **(c)** distance travelled per unit time | Speed = distance/time. |
| 3 | **(d)** odometer | Shows kilometres covered. |
| 4 | **(d)** 3600 seconds | 60 x 60. |
| 5 | **(c)** metre per second | m/s. |
| 6 | **(d)** speedometer | Shows km/h. |
| 7 | **(b)** time period | Time for one to-and-fro swing. |
| 8 | **(c)** kilometre per hour | km/h. |
| 9 | **(c)** distance-time graph | Plots distance vs time. |
| 10 | **(c)** oscillation | Bob returns to the starting point. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Distance travelled per unit time (1); m/s (1). | 1+1 |
| 12 | 400/80 (1); 5 m/s (1). | 1+1 |
| 13 | Uniform: equal distances in equal time (1); Non-uniform: unequal (1). | 1+1 |
| 14 | Time for one complete oscillation (1); Measured in seconds (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | T = 60/30 = 2 s (1); Oscillations in 1 minute = 30 (1); Time for one oscillation is 2 s (1). | 1+1+1 |
| 16 | 72 x 5/18 = 20 m/s (1); 15 x 18/5 (1); 54 km/h (1). | 1+1+1 |
| 17 | 20 min = 1/3 h (1); 18 x 1/3 (1); 6 km (1). | 1+1+1 |
| | **OR:** 180/4 = 45 km/h (1); 45 x 5/18 (1); 12.5 m/s (1). | 1+1+1 |
| 18 | 400/5 = 80 km/h (1); 80 x 3 (1); 240 km (1). | 1+1+1 |
| | **OR:** Speedometer shows speed (1); Odometer shows distance (1); Helps check time and fuel (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Hang pendulums of lengths 25 cm and 100 cm (1); Time 20 oscillations each (1); Find T = t/20 (1); Longer has greater period (1); Mass has no effect (1). | 1+1+1+1+1 |
| | **OR:** Oscillation: one to-and-fro (1); Time period: time for one (1); Amplitude: extreme displacement (1); Bob returns (1); Periodic motion (1). | 1+1+1+1+1 |
| 20 | First 60 x 2 = 120 km (1); Second 40 x 3 = 120 km (1); Total 240 km in 5 h (1); Average 240/5 (1); 48 km/h (1). | 1+1+1+1+1 |
| | **OR:** Cyclist 5 m/s (1); Car 25 m/s (1); Snail 0.005 m/s (1); Arrange in order (1); Car fastest (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Dividing time by the number of oscillations the wrong way; time period = **total time ÷ number of oscillations**.
- Writing that a heavier bob has a larger time period; the period depends on **length**, not on mass.
- Forgetting to convert units before calculating (minutes to hours, km/h to m/s).
- Using $\times \frac{18}{5}$ instead of $\times \frac{5}{18}$ when converting km/h to m/s.
- Calculating average speed as the average of two speeds instead of *total distance divided by total time*.
- Reading a horizontal line on a distance-time graph as 'moving fast'; it means the object is **at rest**.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** distance / time | Speed. |
| 2 | **(c)** straight line | Constant speed gives a straight line. |
| 3 | **(d)** length | Longer pendulum, longer period. |
| 4 | **(b)** 60 km/h | 120/2. |
| 5 | **(b)** at rest | Distance does not change. |
| 6 | **(c)** 2 s | 40/20. |
| 7 | **(c)** 5 m/s | 100/20. |
| 8 | **(d)** higher speed | More distance per time. |
| 9 | **(d)** the shadow of a rod | Shadow moves with the Sun. |
| 10 | **(d)** 180 km | 60 x 3. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | 54 x 5/18 (1); 15 m/s (1). | 1+1 |
| 12 | Sundial (1); Water clock (1). | 1+1 |
| 13 | One complete to-and-fro motion (1); Back to starting point (1). | 1+1 |
| 14 | Graph of distance against time (1); Shows motion (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Hang a bob from a thread (1); Count 20 oscillations with a stopwatch (1); Divide the time by 20 (1). | 1+1+1 |
| 16 | Shadow of rod changes (1); Shows time of day (1); No sunlight at night (1). | 1+1+1 |
| 17 | Distance not proportional to time (1); Speed increases (1); Non-uniform (1). | 1+1+1 |
| | **OR:** Reduces error in starting and stopping (1); Larger time measured (1); Divide to get one (1). | 1+1+1 |
| 18 | Water flows at a steady rate (1); Level marks time (1); Measures fixed durations (1). | 1+1+1 |
| | **OR:** It increases (1); Longer pendulum swings slower (1); Mass does not matter (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Sundial uses shadows (1); Water clock (1); Sand clock (1); Pendulum clock (1); Quartz watch (1). | 1+1+1+1+1 |
| | **OR:** Reset to zero (1); Start at the event (1); Stop at the end (1); Avoid reaction delay (1); Repeat and average (1). | 1+1+1+1+1 |
| 20 | Speed = distance / time (1); Uniform: straight line graph (1); Non-uniform: curved graph (1); Example: highway car (1); Example: bus in traffic (1). | 1+1+1+1+1 |
| | **OR:** 1 h = 60 min (1); 1 min = 60 s (1); 1 h = 3600 s (1); Common standard (1); Universal clocks (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Dividing time by the number of oscillations the wrong way; time period = **total time ÷ number of oscillations**.
- Writing that a heavier bob has a larger time period; the period depends on **length**, not on mass.
- Forgetting to convert units before calculating (minutes to hours, km/h to m/s).
- Using $\times \frac{18}{5}$ instead of $\times \frac{5}{18}$ when converting km/h to m/s.
- Calculating average speed as the average of two speeds instead of *total distance divided by total time*.
- Reading a horizontal line on a distance-time graph as 'moving fast'; it means the object is **at rest**.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** 5/18 m/s | 1000/3600. |
| 2 | **(c)** water clock | Clepsydra. |
| 3 | **(c)** 5 h | 200/40. |
| 4 | **(d)** 10 m/s | 36 x 5/18. |
| 5 | **(d)** quartz | Quartz crystals vibrate regularly. |
| 6 | **(c)** uniform | Constant speed. |
| 7 | **(b)** 72 km/h | 20 x 18/5. |
| 8 | **(d)** swinging of a pendulum | Repeats regularly. |
| 9 | **(b)** non-uniform | Speed changes. |
| 10 | **(a)** m/s | Metre per second. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Its swings take equal time (1); Gives regular timing (1). | 1+1 |
| 12 | Speed = distance / time (1); 150 / 3 = 50 km/h (1). | 1+1 |
| 13 | Odometer: distance (1); Speedometer: speed (1). | 1+1 |
| 14 | Time = distance / speed (1); 90/45 = 2 h (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Uniform: equal distances in equal times (1); Non-uniform: not equal (1); Example: car on a highway vs in traffic (1). | 1+1+1 |
| 16 | Quartz crystal vibrates at a fixed rate (1); Counted by a circuit (1); Very accurate (1). | 1+1+1 |
| 17 | Sports (1); Railways (1); Experiments (1). | 1+1+1 |
| | **OR:** Straight line (1); Through origin (1); Slope shows speed (1). | 1+1+1 |
| 18 | One round in 60 s (1); So 60 s (1); Periodic motion (1). | 1+1+1 |
| | **OR:** Total distance 100 km (1); Total time 3 h (1); Average 33.3 km/h (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Equal distances in equal times (1); Uniform motion (1); Speed = 10/2 (1); 5 m/s (1); Graph is straight (1). | 1+1+1+1+1 |
| | **OR:** Distance 45 x 4 = 180 km (1); Time = 180/60 (1); 3 h (1); Speed up reduces time (1); Inversely proportional (1). | 1+1+1+1+1 |
| 20 | Odometer: distance (1); Speedometer: speed (1); Odometer from 12000 to 12120 km (1); Time 2 h (1); Average 60 km/h (1). | 1+1+1+1+1 |
| | **OR:** Moon's revolution (1); Pendulum (1); Heartbeat (1); Non-periodic: falling stone (1); Flying bird (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

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